

Recurrent Neural Networks

Suited for sequential data

Consider a data point to be of following nature

$x = \{x^1, x^2, \dots, x^T\}$: single data point.

where $x^t \in \mathbb{R}^d$

Eg : natural language
clinical time-series
video

Example tasks on such data:

i) sequence classification
emotion extraction from text.

ii) sequence to sequence regression.
machine translation.

In sequential data, every datapoint may be of different length. & thus MLP cannot handle them.

RNNs

parameter sharing across "time"

$$x = \{x_1, x_2, \dots, x_T\}$$

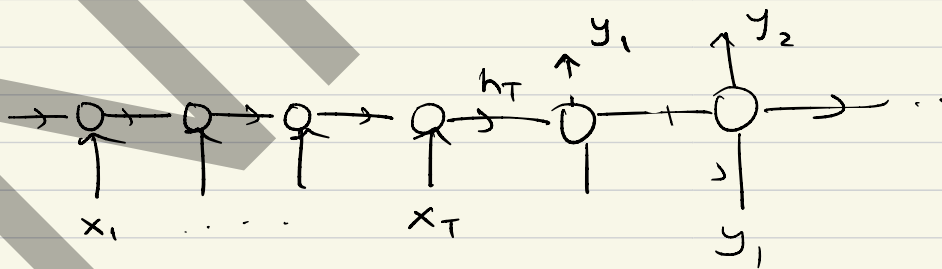
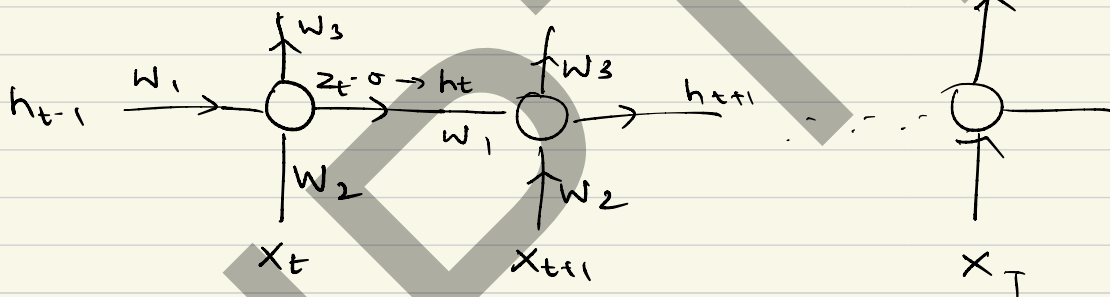
$$z_t = W_1 h_{t-1} + W_2 x_t + b_1$$

$$h_t = \sigma(z_t)$$

$$\hat{y}_t = W_3 h_t + b_2$$

single RNN cell

$$\hat{y} \quad L(\hat{y}, y)$$



Back-propagation in RNNs

BPTT leads to vanishing gradients in long sequences.

vanishing gradients in RNNs.

At each time step t ,

$$z_t = W_1 h_{t-1} + W_2 x_t$$

$$h_t = \sigma(z_t)$$

suppose L_T be the loss at time T .

$$\frac{\partial L_T}{\partial h_t} = \frac{\partial L_T}{\partial h_T} \frac{\partial h_T}{\partial h_t}$$

Consider $\frac{\partial h_T}{\partial h_t} = \prod_{k=t}^{T-1} \frac{\partial h_{k+1}}{\partial h_k}$

Consider

$$\frac{\partial h_{k+1}}{\partial h_k}$$

we have $h_{k+1} = \sigma(W_1 h_k + W_2 x_{k+1})$

$$\frac{\partial h_{k+1}}{\partial h_k} = \text{diag}(\sigma'(z_{k+1})) W_1.$$

consider the norm of the derivative,

$$\left\| \frac{\partial h_T}{\partial h_t} \right\| \leq \prod_{k=t}^{T-1} \left[\left\| \text{diag}(\sigma'(z_{k+1})) \right\| \cdot \|w_1\| \right]$$

$$\left\| \text{diag}(\sigma'(z_{k+1})) \right\| \leq 1$$

Let $\alpha_{\max}$ be the largest singular value of w_1 , $\|w_1\| = \alpha_{\max}$

$$\Rightarrow \left\| \frac{\partial h_T}{\partial h_t} \right\| \leq (\alpha_{\max})^{T-t}$$

as the gap $T-t$ increases, $(\alpha_{\max})^{T-t}$ decays exponentially.

The General Mathematical objective.

In a recurrent system,

$$h_t = f(h_{t-1}, x_t)$$

$$\Rightarrow \frac{\partial h_T}{\partial h_t} = \prod_{k=t}^{T-1} \frac{\partial h_{k+1}}{\partial h_k}$$

In a vanilla RNN, vanishing grad happens since $\frac{\partial h_{k+1}}{\partial h_k} = \text{diag}(\sigma'(1)) \cdot W$

solution is to make $\frac{\partial h_{k+1}}{\partial h_k} \approx I$

The state update should be additive.

$$h_t = \alpha_t \odot h_{t-1} + \beta_t \odot \tilde{h}_t$$

where h_{t-1} is the previous state

$$\tilde{h}_t = \sigma(W_2 x_t + W_1 h_{t-1})$$

α_t & β_t are "modulators"

The gradients for the above networks.

Consider

$$\frac{\partial h_t}{\partial h_{t-1}} = \text{diag}(\alpha_t) + \frac{\partial \alpha_t}{\partial h_{t-1}} \text{diag}(h_{t-1}) + \frac{\partial (\beta_t \odot \tilde{h}_t)}{\partial h_{t-1}} \quad \left. \vphantom{\frac{\partial h_t}{\partial h_{t-1}}} \right\} \varepsilon_t$$

$$= \text{diag}(\alpha_t) + \Sigma_t(w_1, w_2)$$

$$\frac{\partial h_T}{\partial h_t} = \prod_{k=t}^{T-1} \left(\text{diag}(\alpha_{k+1}) + \Sigma_{k+1} \right)$$

$$\left\| \frac{\partial h_T}{\partial h_t} \right\| \leq \prod_{k=t}^{T-1} (\mathbb{I}) = 1$$

$$y = x + F(x) \quad \therefore \text{Residual / skip Connection}$$

